

Claim Safe

Auto Insurance Premium Recommendation System

Pranav Reddy Thummala : G01389371

Shashank Ala : G01382819

Praneeth Boinpally : G01409458

Meghna Peruri : G01394041

Niharika Upadhayula : G01391129

Contributions

1. Shashank Ala(G01382819):

Data Collection and Integration:

- Responsible for gathering and integrating data from Carfax, claims history, credit history, and driver data.
- Ensured the integrity and accuracy of data inputs that feed into the risk assessment model.
- Maintained and updated data repositories as new data becomes available.

2. Pranav Reddy Thummala(G01389371):

Developed Risk Assessment Model:

- Developed and refined the Risk Assessment Model using the collected data.
- Analyzed different risk factors from vehicle history to driver behavior and financial reliability.
- Produced a quantified risk score that assesses the potential risk associated with insuring each driver.

3. Praneeth Boinpally(G01409458):

Developed Base price Calculation Model:

- Used company data to build and manage the Base Price Calculation Model.
- Calculated base prices by considering operational costs, claim costs, and targeted profit margins.
- Ensured that the base price output accurately reflects the company's financial strategy and market competitiveness.

4. Meghna Peruri(G01394041):

Developed Premium Calculation Model:

- Integrated outputs from both the Risk Assessment Model and the Base Price Calculation Model.
- Adjusted the base insurance premiums according to the risk scores to derive the final premium rates.
- Applied analytical methods to accommodate various discounting strategies based on driver eligibility, ensuring premiums calculated are competitive and reflective of actual risk.

5. Niharika Upadhayula(G01391129):

Output Management and Reporting:

- Produced final premium calculation outputs and detailed insurance premiums for different coverage types.
- Analyzed model performance, recommending enhancements based on new data and market

Competition.

Collaborative Duties:

Cross-Verification: We periodically reviewed each other's work to ensure accuracy and coherence in the data flow and model outputs.

Documentation and Communication: All of us contributed to the documentation process, detailing methodologies, data sources, and model insights.

Regular Updates and Meetings: We met regularly to update each other on progress, challenges, and insights, ensuring cohesive project advancement.

Table of Contents

1. Abstract

- Overview of the Project
- Problem Addressed
- Summary of Contributions

2. Introduction

- Importance of the Problem Domain
- High-Level Description of the Decision Problem
- Review of Existing Research and Limitations
- Summary of Key Contributions

3. System Functionality

- User Interaction with the System
- Example of System Use
- Benefits to the User

4. System High-Level Architecture

- Diagram of System Architecture
- Description of Data Flow

5. Analytic Model

- Description of Inputs
- Description of Model
- Description of Outputs

6. Conclusion and Future Work

- Conclusion
- Future Work

Abstract

1. Overview of the Project:

The project develops an advanced auto insurance recommendation system tailored for insurance companies. This system is designed to calculate optimal insurance premiums for individual drivers by analyzing a comprehensive set of factors. These factors include the driver's historical data, such as driving history, past traffic violations and accidents, claims history, and payment records. Additionally, personal attributes like age and location, along with detailed vehicle information obtained from Carfax reports, are integrated to assess each driver's risk profile comprehensively.

The model not only evaluates the driver's risk but also incorporates the insurance company's operational perspectives, such as the cost of operations, estimated claims expenses, and the customer base's volume. Moreover, it strategically considers the prevailing market prices and competition to ensure that the recommended premiums are not only risk-appropriate but also competitive and sustainable in the market. Importantly, the system also factors in the profit margins set by the insurance company, ensuring that the pricing strategy supports the company's financial goals and sustainability.

By balancing these elements, the system determines the most suitable premium for various insurance packages, including options like collision only, collision plus liability, and comprehensive coverage that includes collision, liability, and additional risks.

This innovative system aims to enable insurance companies to offer dynamically priced insurance policies that reflect actual risk, improve customer satisfaction through fair and transparent pricing, and enhance operational efficiency by aligning premium pricing more closely with anticipated claims costs, market dynamics, and profitability requirements.

2. Problem Addressed:

The auto insurance industry faces significant challenges in accurately pricing premiums that reflect the true risk associated with individual drivers. Traditional methods often result in broad categorizations, which can lead to inequitable premium settings where low-risk drivers may subsidize high-risk drivers, or vice versa. This inefficiency not only affects customer satisfaction but also impacts the financial performance of insurance companies.

The recommendation system developed in this project addresses these challenges by enabling insurance companies to precisely assess the risk associated with each driver. By integrating a wide range of factors from driving records and vehicle history to broader demographic data, the system provides a nuanced risk profile for each individual. This approach allows for the setting of premiums that are truly reflective of each driver's risk, rather than applying a one-size-fits-all model.

Moreover, the system takes into account various business factors on the company's side, such as operational costs, anticipated claim expenses, and market competition. By doing so, it ensures that the recommended premiums are not only equitable and risk-appropriate but also competitive and profitable. This strategic pricing capability empowers insurance companies to thrive in a highly competitive industry by optimizing their rate structures, thereby enhancing their profitability and market positioning.

This problem-solving approach represents a significant advancement in the way auto insurance premiums are determined, moving towards more personalized, fair, and financially sustainable practices.

3. Summary of Contributions

Our team has made significant contributions to this project, achieving key milestones that together have driven our solution forward. We developed a complex analytic model that combines and processes a variety of information, including driving records, vehicle details, and personal data. We also carried out thorough data collection and management to make sure our model used accurate and relevant data. Additionally, we factored in important business elements such as operational costs and profit margins to help create pricing strategies that are sustainable. We conducted extensive testing and fine-tuning to ensure our model works well in various situations, enabling insurance companies to set fair and accurate insurance premiums. These efforts show our dedication to pioneering improvements in how auto insurance premiums are calculated.

Introduction

1. Importance of the Problem Domain

The domain of auto insurance premium determination is critically important as it directly impacts both the financial stability of insurance companies and the affordability of policies for consumers. Accurately setting premiums is essential because it balances the risk insurers take with the rates charged to customers. If premiums are set too high, insurance becomes unaffordable for many drivers; set them too low, and insurers may suffer unsustainable losses from claims. Furthermore, with advancements in technology and data analytics, consumers expect more personalized and fair pricing based on their specific circumstances and risk profiles. This demand for precision in premium setting underscores the need for sophisticated models that can accurately assess individual risk using a variety of factors. By improving how premiums are calculated, insurance companies can enhance customer satisfaction, reduce financial risks, and maintain competitive advantages in a highly competitive market. This underscores the significance of developing innovative solutions in this area, which can lead to more equitable insurance practices and better financial outcomes for both insurers and insureds.

2. High-Level Description of the Decision Problem

The decision problem at the center of this project involves figuring out the best auto insurance premiums for individual drivers—a critical issue that affects both insurance companies and their customers. Insurance companies must set premiums that accurately reflect the risk each driver poses without unfairly pricing policies or losing customers to competitors with better rates.

At its heart, the decision problem is complex and requires analyzing a wide range of data that affects a driver's risk level. This data includes driving records, claims history, vehicle details, and personal demographics. The decision system also needs to consider external factors like market competition and regulatory rules, which can differ greatly from one place to another and over time.

The decision-making process is made even more challenging by the changing nature of risk factors. For example, a driver's risk profile might change due to new traffic violations or changes in their lifestyle that could influence their driving habits. As a result, insurance companies need a decision guidance system that can manage not just historical data but also adjust to these changing risk factors both effectively and accurately.

Additionally, the solution needs to support the company's strategic objectives, such as staying profitable, reducing claim payouts, and improving customer satisfaction with fair pricing. These goals require a decision system that balances risk assessment with business needs, ensuring that premium setting is sensitive to risk and financially sound.

3. Review of Existing Research and Limitations

The current existing systems for auto insurance recommendations are designed to handle the specific challenges of the auto insurance sector, where accuracy and trust are crucial because of the significant impact of each customer interaction. However, these systems face several limitations.

One major limitation is the need to adapt general recommendation technologies to the specific requirements of auto insurance. This industry has a relatively small set of product options, different types of coverage which demands highly accurate recommendations. Furthermore, the complexity of insurance products, which may include overlapping coverages and detailed eligibility criteria, adds another layer of challenge.

These complexities mean that typical recommendation algorithms used in broader commercial sectors often fall short without significant adjustments. Therefore, auto insurance recommendation systems must be finely tailored to effectively match insurance products with customer needs, ensuring recommendations are both practical and trustworthy.

4. Summary of Key Contributions

Our project has made significant strides in improving the effectiveness and reliability of auto insurance recommendation systems. Initially, we conceptualized the problem by identifying and defining the unique challenges associated with accurately recommending auto insurance premiums. Understanding the necessity for precise and individualized risk assessments allowed us to focus on the most relevant issues in the auto insurance domain.

We then developed a formal analytic model using Python, designed to process a wide array of data inputs such as driving history, vehicle information, and personal demographics. This sophisticated model is capable of assessing risk with high accuracy, which is critical for determining appropriate insurance premiums.

Following the model development, we implemented a system that makes optimal recommendations for insurance premiums. This system not only considers the risk associated with individual drivers but also integrates broader business factors including operational costs, claims expenses, and competitive market pricing. This ensures that the recommendations are not only appropriate from a risk perspective but also economically viable and competitive in the market.

To demonstrate the practical application of our system, we exemplified it using a use-case. This real-world example highlighted how the system can be employed to provide tailored insurance recommendations, significantly enhancing customer satisfaction and profitability for insurance companies. Through these contributions, we have significantly advanced the use of decision making algorithms in crafting more precise and effective auto insurance premium recommendations, aligning them closely with both consumer needs and business objectives.

System Functionality

1. User Interaction with the System

Our auto insurance recommendation system is designed to simplify and optimize the process by which insurance companies set premiums based on detailed risk assessments. Here's a general overview of how an insurance company can utilize our system effectively:

Inputting Data

An insurance company starts by entering essential business information into the system, including operational costs, claim costs, profit margins, and details about the types of insurance services offered, such as liability, collision, and comprehensive insurance. Additionally, the system requires detailed client data for accurate risk assessment. This includes personal demographics, driving history, vehicle details, past violations, accidents, and any other relevant information.

Using the System

Once the data is inputted, the system processes this information to calculate risk scores for each driver, reflecting their specific circumstances and risk factors. These risk scores are then used to recommend optimal premium rates for various insurance coverages. The recommendations are formulated considering both the insurance company's business factors, such as profitability and market competitiveness, and the individual risk factors, such as the likelihood of claims based on the driver's history and vehicle condition.

Reviewing Recommendations

The insurance company can review the system-generated recommendations, which include detailed premium rates and risk scores for each client. The system provides a clear breakdown of how each premium is calculated, allowing the company to make informed decisions about adjusting rates or applying discounts. This functionality enables the company to ensure that the premiums are not only competitive and financially sound but also fair and closely aligned with the actual risk each client presents.

Regular Updates and Monitoring

The system is designed to accommodate regular updates to client profiles, ensuring that premium recommendations remain accurate as new data becomes available or as existing data changes, such as a client receiving a new traffic violation or updating their vehicle's safety features. This dynamic feature of the system helps the insurance company maintain up-to-date risk assessments and pricing strategies.

2. Example of system use

Let's take the example of a hypothetical insurance company, InsuraMax, to illustrate how they can effectively utilize our auto insurance recommendation system to streamline their premium setting process.

Inputting Data

InsuraMax starts by entering crucial company metrics into the system. These include:

Operational costs: \$350,000

Claim costs: \$170,000

Profit margin: 10%

Expected customer volume: 1,000

Additionally, they provide details about their insurance services, such as liability, collision+liability, and comprehensive coverage options. On the client side, InsuraMax inputs comprehensive data for each driver, including demographics, driving history, vehicle details, and records of any violations or accidents.

Using the System

Using the data provided, the system calculates individual risk scores for drivers based on their specific profiles. For example, it considers factors such as:

Risk Score: Risk score is calculated for each driver based on various factors such as his driving history, payment history, past claims history etc.

Vehicle use and mileage: Such as 15,000 annual miles for a personal vehicle.

Safety features: Like Automatic Emergency Braking and Electronic Stability Control, which can mitigate risk.

The system then uses these risk assessments to recommend optimal premium rates across various insurance products. For instance, it might suggest a premium of \$95.30 for liability coverage, \$132.75 for collision+liability, and \$157.30 for comprehensive coverage for a driver based on their risk score and the business metrics of InsuraMax.

Reviewing Recommendations

InsuraMax reviews the system-generated recommendations, which detail the premiums and risk scores for each type of coverage. The detailed breakdown allows InsuraMax to see how factors like operational costs, expected claim costs, and competitive market conditions influence the premium calculations. If necessary, InsuraMax can adjust these recommendations to apply discounts or adjust rates based on additional internal assessments or changes in the market.

3. Benefits to user

Our auto insurance recommendation system offers a multitude of benefits to insurance companies, streamlining their operations and enhancing their service offerings in several critical ways:

Accurate Risk Assessment

The system provides precise risk assessments by analyzing a comprehensive array of data from individual drivers and their vehicles. It considers factors such as driving records, accident histories, vehicle safety features,

and personal demographics to ensure that each risk calculation accurately reflects the true risk posed by each driver. This accuracy is crucial for setting fair and appropriate premiums.

Dynamic Pricing Strategy

With its advanced analytics capabilities, the system allows for dynamic pricing strategies. Insurance companies can adjust premiums in real-time based on updated data such as changes in driving behavior, modifications to vehicles, or new traffic violations. This adaptability helps insurers offer rates that are not only competitive but also reflective of the most current risk assessments.

Enhanced Operational Efficiency

By automating the complex processes of data integration and risk calculation, the system streamlines operations, significantly reducing the time and effort required to set premiums. This increase in efficiency allows insurers to allocate more resources towards strategic initiatives and customer engagement, minimizing the scope for human error and enhancing overall productivity.

Improved Financial Performance

The system helps insurance companies optimize their financial performance by aligning premium rates with actual risk while considering business factors like operational costs and profit margins. This alignment ensures that the premiums are financially viable and competitive, helping to improve the company's profitability and maintain financial health.

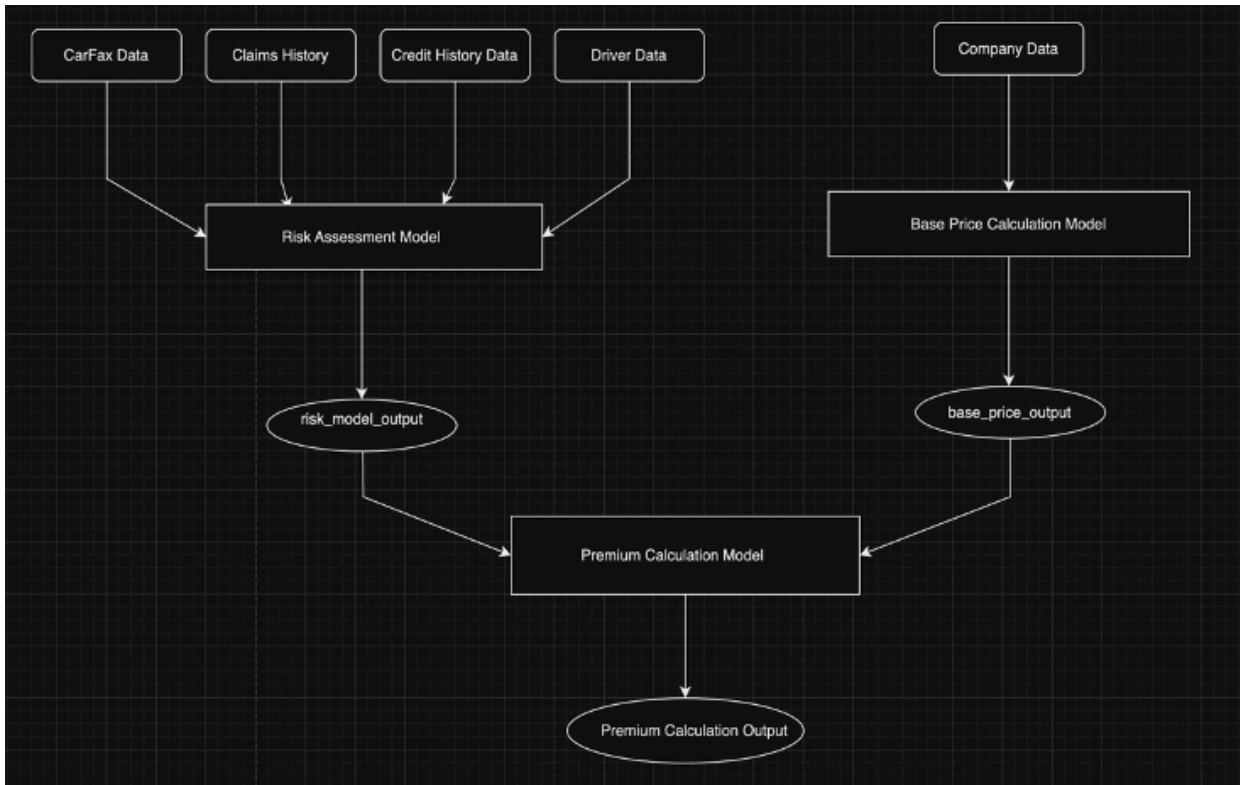
Increased Customer Satisfaction

Customers benefit from fair and transparent premium settings that are directly linked to their specific risk profiles. This clarity and fairness can increase customer trust and satisfaction, making them more likely to remain with the insurer and potentially leading to higher retention rates.

Overall, our auto insurance recommendation system enhances the capabilities of insurance companies, enabling them to operate more effectively and efficiently while providing high-quality, tailored services to their customers.

System Architecture

Flow:



Data Inputs:

The system begins with multiple sources of input data:

CarFax Data: Provides detailed reports on individual vehicles, such as accident history, service history, and previous ownership.

Claims History: Contains data about past claims made by drivers, which can include the frequency and cost of those claims.

Credit History Data: Involves the financial history of drivers, including credit scores and payment histories.

Driver Data: Information about the drivers themselves, including age, driving record, and other personal details.

Company Data: This includes broader company-specific data like operational costs, expected customer volume, and strategic financial metrics such as profit margins.

Risk Assessment Model:

All relevant data from CarFax, claims history, credit history, and driver data feeds into the Risk Assessment Model.

This model analyzes the input data to assess the risk level associated with insuring each driver, based on their vehicle condition, driving behavior, and financial reliability.

The output from this model is a **risk model output**, which quantifies the risk in a format that can be used to adjust insurance pricing.

Base Price Calculation Model:

Utilizes company data to calculate a base price for insurance policies.

This calculation considers the operational and claim costs of the company, incorporating desired profit margins to ensure the financial sustainability of the insurance offerings.

The output here is a **base price output** which provides a foundational cost before individual risk factors are considered.

Premium Calculation Model:

This model receives two key pieces of input: the risk model output from the Risk Assessment Model and the base price output from the Base Price Calculation Model.

It integrates these inputs to determine the final insurance premiums for different types of coverage. This involves adjusting the base price according to the quantified risk associated with each driver, ensuring that premiums are proportionate to the potential cost of claims.

The premiums are calculated to reflect both the risk and the necessary return on investment per the company's financial targets.

Output:

The final output of the system is the premium calculation output, which details the specific premium costs for each insurance policy. This output considers the inherent risk, the cost of potential claims, and the company's financial requirements.

Analytic Model

1. Description of Inputs:

· Driver.json:

This input provides a comprehensive dataset on an individual driver, crucial for the auto insurance recommendation system to assess risk and determine appropriate insurance premiums. Here's a breakdown of the components of this input.

Driver Information

Includes the driver's name, address, date of birth, driver license number, and license class. This basic information is essential for identifying the driver within the system and ensuring that all records are accurately linked to the correct individual.

License Status

Provides the current status of the driver's license (active or otherwise) and its expiration date. An active status is critical for verifying that the driver is legally permitted to drive.

Violations

Traffic Violations

Lists specific instances of driving violations, such as speeding or failing to obey traffic signs, along with associated points and the duration these points affect the driver's record. This section is particularly important for assessing driving behavior and risk propensity.

Accidents

Details past accidents, specifying the type, a brief description, fault status, points incurred, and the duration of these points. Accident history helps in evaluating the driver's involvement in incidents and their potential risk factor.

Suspensions

Records any suspensions of the driver's license, including reasons for the suspension, reinstatement dates, associated points, and the duration of these points.

Safe Driving Points:

Reflects points awarded for safe driving practices, which can be used to offset points from violations or accidents, indicating a measure of the driver's overall safety record.

Vehicle Information

Provides the vehicle identification number (VIN), make, model, year, primary use, annual mileage, and safety features such as Automatic Emergency Braking, Electronic Stability Control, and Adaptive Headlights. These elements are vital for assessing the safety and risk associated with the vehicle itself.

Insurance and Location Details

The number of years the vehicle has been insured, which can be indicative of the driver's stability and reliability. Includes type of location (urban or rural), population density, and crime rate, which are all factors that can influence risk levels due to differing accident rates and potential for vehicle theft or damage in different areas.

```
{
  "driver_information": {
    "name": "John Doe",
    "address": "123 Main St, Anytown, VA, 12345",
    "date_of_birth": "1985-06-15",
    "driver_license_number": "VA123456789",
    "license_class": "Class D"
  },
  "license_status": {
    "status": "Active",
    "expiration_date": "2029-06-15"
  },
  "violations": [
    {
      "date": "2020-05-10",
      "violation": "Speeding 10-14 mph above the posted speed limit",
      "points": 4,
      "duration": "5 years"
    },
    {
      "date": "2021-03-15",
      "violation": "Failure to obey highway sign",
      "points": 3,
      "duration": "3 years"
    }
  ],
  "accidents": [
    {
      "date": "2019-07-08",
      "type": "Minor",
      "description": "Rear-end collision, minor damage",
      "at_fault": "Yes",
      "points": 3,
      "duration": "3 years"
    }
  ]
},
```

```
  "history": {
    "suspensions": [
      {
        "date": "2018-01-01",
        "reason": "Driving under the influence",
        "reinstated_date": "2018-07-01",
        "points": 6,
        "duration": "11 years"
      }
    ],
    "safe_driving_points": {
      "total_safe_points": 2
    }
  },
  "vehicle_information": {
    "vin": "1HGCM82633A004352",
    "make": "Honda",
    "model": "Civic",
    "year": 2018,
    "vehicle_use": "Personal",
    "annual_mileage": 15000,
    "safety_features": [
      "Automatic Emergency Braking",
      "Electronic Stability Control",
      "Adaptive Headlights"
    ]
  },
  "insured_for": 3,
  "location": {
    "type": "Urban",
    "population_density": "High",
    "crime_rate": "Moderate"
  }
}
```

Similarly, we have 10 driver details in our data set.

Credit_history.json:

Driver License Number

Acts as a unique identifier for each driver, linking their financial data to their driving records. This ensures that the financial information corresponds to the correct individual in the system.

Social Security Number (SSN)

While only partially shown for privacy reasons (represented as XXX-XX-XXXX), the SSN serves as an additional identifier that may be used internally for verifying the identity of the driver and facilitating credit checks.

Credit Score

A numerical value representing the creditworthiness of the driver. A higher credit score indicates better financial stability and reliability, which is a positive factor in risk assessment for insurance purposes. Credit scores are a crucial element as they can influence the terms of the insurance policy, including premium rates.

Monthly Payments on Time

This boolean indicator shows whether the driver consistently makes their monthly payments on time. A true value signifies a positive payment history, suggesting financial reliability, while a false value may indicate potential financial risk.

```
[
  {
    "driver_license_number": "VA123456789",
    "ssn": "XXX-XX-8164",
    "credit_score": 750,
    "monthly_payments_on_time": true
  },
  {
    "driver_license_number": "VA987654321",
    "ssn": "XXX-XX-2143",
    "credit_score": 700,
    "monthly_payments_on_time": true
  },
  {
    "driver_license_number": "VA112233445",
    "ssn": "XXX-XX-8164",
    "credit_score": 670,
    "monthly_payments_on_time": false
  },
  {
    "driver_license_number": "VA556677889",
    "ssn": "XXX-XX-0153",
    "credit_score": 710,
    "monthly_payments_on_time": true
  },
  {
    "driver_license_number": "VA998877665",
    "ssn": "XXX-XX-1256",
    "credit_score": 690,
    "monthly_payments_on_time": false
  },
  {
    "driver_license_number": "VA223344556",
    "ssn": "XXX-XX-2000",
    "credit_score": 770,
    "monthly_payments_on_time": true
  },
  {
    "driver_license_number": "VA667788990",
    "ssn": "XXX-XX-8900",
    "credit_score": 645,
    "monthly_payments_on_time": false
  },
  {
    "driver_license_number": "VA445566778",
    "ssn": "XXX-XX-3425",
    "credit_score": 714,
    "monthly_payments_on_time": true
  },
  {
    "driver_license_number": "VA334455667",
    "ssn": "XXX-XX-6734",
    "credit_score": 689,
    "monthly_payments_on_time": false
  },
  {
    "driver_license_number": "VA778899001",
    "ssn": "XXX-XX-5654",
    "credit_score": 609,
    "monthly_payments_on_time": false
  }
]
```

This financial dataset is used for assessing risk by correlating driver's credit scores and payment histories with their financial reliability. Insurers adjust premiums based on these financial behaviors, increasing rates for those with poor credit or offering discounts to those who consistently make timely payments. This approach allows

for customized insurance offers, enhancing risk management and tailoring policies to individual financial profiles.

· Claims_history.json:

Driver License Number

Serves as a unique identifier for each driver, linking their claims history to their specific profile within the insurance system.

Previous Claims:

Minor Claims: Indicates the number of minor claims the driver has made. Minor claims typically involve less severe damage or incidents that don't result in significant financial loss to the insurer.

Total Loss: Refers to the number of claims that resulted in a total loss, which generally means the vehicle was declared a total write-off due to the extent of damage or the cost of repair exceeding the value of the vehicle.

```
[
  {
    "driver_license_number": "VA123456789",
    "previous_claims": {
      "minor_claims": 1,
      "total_loss": 1
    }
  },
  {
    "driver_license_number": "VA987654321",
    "previous_claims": {
      "minor_claims": 0,
      "total_loss": 0
    }
  },
  {
    "driver_license_number": "VA112233445",
    "previous_claims": {
      "minor_claims": 1,
      "total_loss": 0
    }
  },
  {
    "driver_license_number": "VA556677889",
    "previous_claims": {
      "minor_claims": 1,
      "total_loss": 0
    }
  },
  {
    "driver_license_number": "VA998877665",
    "previous_claims": {
      "minor_claims": 0,
      "total_loss": 0
    }
  }
],
[
  {
    "driver_license_number": "VA223344556",
    "previous_claims": {
      "minor_claims": 2,
      "total_loss": 0
    }
  },
  {
    "driver_license_number": "VA667788990",
    "previous_claims": {
      "minor_claims": 4,
      "total_loss": 0
    }
  },
  {
    "driver_license_number": "VA445566778",
    "previous_claims": {
      "minor_claims": 0,
      "total_loss": 0
    }
  },
  {
    "driver_license_number": "VA334455667",
    "previous_claims": {
      "minor_claims": 0,
      "total_loss": 0
    }
  },
  {
    "driver_license_number": "VA778899001",
    "previous_claims": {
      "minor_claims": 0,
      "total_loss": 0
    }
  }
]
```


This dataset, detailing driver's previous minor and total loss claims, is essential for evaluating insurance risk. A higher number of past claims, particularly total losses, suggests increased risk, potentially raising premium costs. Insurers use this data for precise premium calculations and to tailor policies that match each driver's risk profile, enhancing the accuracy and customization of insurance offerings.

· **Carfax_report.json:**

VIN: Each vehicle's unique identifier.

Make and Model: Specifies the manufacturer and model of the vehicle.

Year: The year the vehicle was manufactured.

Title Status: Indicates the legal status of the vehicle's title (e.g., Clean, Rebuilt), which affects its insurability and value.

Ownership History:

Number of Owners: Indicates how many times the vehicle has changed hands.

Purchase Type: Specifies whether the vehicle was purchased for personal or commercial use.

Accident History:

Minor Accidents: The count of minor accidents the vehicle has been involved in.

Major Accidents: The count of major accidents that resulted in significant damage.

Service History:

Service Records: The number of times the vehicle has been serviced, which can indicate how well the vehicle has been maintained.

Repair History:

Minor Repairs: Number of minor repairs the vehicle has undergone.

Major Repairs: Number of significant repairs, which could include engine or transmission rebuilds.

Open Recalls: The number of unresolved manufacturer recalls on the vehicle, indicating potential safety risks.

Current Mileage: The most recent recorded mileage, which provides insight into the vehicle's usage and wear.

```

{
  "vin": "3N1AB7AP6KY237745",
  "make": "Nissan",
  "model": "Sentra",
  "year": 2019,
  "title_status": "Rebuilt",
  "ownership_history": {
    "number_of_owners": 3,
    "purchase_type": "Personal"
  },
  "accident_history": {
    "minor_accidents": 2,
    "major_accidents": 1
  },
  "service_history": {
    "service_records": 7
  },
  "repair_history": {
    "minor_repairs": 2,
    "major_repairs": 3
  },
  "open_recalls": {
    "number_of_open_recalls": 0
  },
  "current_mileage": {
    "date": "2024-04-26",
    "mileage": 80000
  }
},
{
  "vin": "3FA6P0K92HR294653",
  "make": "Ford",
  "model": "Fusion",
  "year": 2017,
  "title_status": "Clean",
  "ownership_history": {
    "number_of_owners": 4,
    "purchase_type": "Lease"
  },
  "accident_history": {
    "minor_accidents": 2,
    "major_accidents": 1
  },
  "service_history": {
    "service_records": 6
  },
  "repair_history": {
    "minor_repairs": 2,
    "major_repairs": 0
  },
  "open_recalls": {
    "number_of_open_recalls": 1
  },
  "current_mileage": {
    "date": "2024-04-26",
    "mileage": 120000
  }
},
{
  "vin": "4S4BSANC9K3284561",
  "make": "Subaru",
  "model": "Outback",
  "year": 2019,
  "title_status": "Clean",
  "ownership_history": {
    "number_of_owners": 1,
    "purchase_type": "Personal"
  },
  "accident_history": {
    "minor_accidents": 1,
    "major_accidents": 0
  },
  "service_history": {
    "service_records": 10
  },
  "repair_history": {
    "minor_repairs": 3,
    "major_repairs": 0
  },
  "open_recalls": {
    "number_of_open_recalls": 0
  },
  "current_mileage": {
    "date": "2024-04-26",
    "mileage": 140000
  }
},

```

Similarly, we have Carfax reports of all the cars of the drivers in our data set.

This dataset aids in vehicle insurance risk assessment by evaluating a vehicle's condition and value based on its title status, accident, and repair history. It helps predict future claim risks from historical damage and repairs and informs premium calculations using service records and mileage. Additionally, it allows insurers to tailor insurance offers based on whether a vehicle is used for personal or commercial purposes, ensuring precise policy customization.

· Car_value.json:

Year: The manufacture year of the vehicle.

Market Value: The current estimated market value of the vehicle.

Repair Costs:

Minor: The cost of minor repairs that the vehicle might need, which generally cover small damages or routine maintenance.

Major: The cost of major repairs that the vehicle might need, typically involving significant damage or major mechanical failures.

```

{
  "Car Models": [
    {
      "Toyota Camry": {
        "Year": 2019,
        "Market Value": 21000,
        "Repair Costs": {
          "Minor": 450,
          "Major": 3900
        }
      },
      "Ford F-150": {
        "Year": 2020,
        "Market Value": 26000,
        "Repair Costs": {
          "Minor": 480,
          "Major": 4100
        }
      }
    ],
    {
      "Chevrolet Silverado": {
        "Year": 2016,
        "Market Value": 23000,
        "Repair Costs": {
          "Minor": 470,
          "Major": 4200
        }
      },
      "Ram 1500": {
        "Year": 2024,
        "Market Value": 28000,
        "Repair Costs": {
          "Minor": 510,
          "Major": 4400
        }
      },
      "Toyota RAV4": {
        "Year": 2017,
        "Market Value": 22000,
        "Repair Costs": {
          "Minor": 430,
          "Major": 4000
        }
      }
    ]
  ],
  "Honda CR-V": {
    "Year": 2019,
    "Market Value": 24000,
    "Repair Costs": {
      "Minor": 460,
      "Major": 4300
    }
  },
  "Chevrolet Equinox": {
    "Year": 2022,
    "Market Value": 19000,
    "Repair Costs": {
      "Minor": 420,
      "Major": 3700
    }
  },
  "Ford Explorer": {
    "Year": 2023,
    "Market Value": 29000,
    "Repair Costs": {
      "Minor": 520,
      "Major": 4600
    }
  },
  "Nissan Rogue": {
    "Year": 2018,
    "Market Value": 18000,
    "Repair Costs": {
      "Minor": 410,
      "Major": 3600
    }
  },
  "Ford Escape": {
    "Year": 2015,
    "Market Value": 15000,
    "Repair Costs": {
      "Minor": 400,
      "Major": 3500
    }
  }
}

```

Similarly we have details of many other vehicles.

This dataset helps insurance companies determine insurance premiums based on vehicle market values, assess risks from potential repair costs, accurately estimate potential claims, and tailor insurance products to accommodate specific repair coverage needs based on the detailed breakdown of repair costs.

· Company.json:

Company Information

Name: InsuraMax

Address: 123 Poplar St, Jersey City, NJ 23675

Contact Details: Phone number, email, and website provided for customer and partner contact.

Services Offered:

InsuraMax offers a range of insurance products including:

Liability Insurance: Covers damages to others caused by the policyholder while operating a vehicle.

Collision + Liability Insurance: Covers damage to the policyholder's car from a collision.

Liability + Collision + Comprehensive Insurance: Extends coverage to include non-collision-related damages to the policyholder's car.

Roadside Assistance: Provides emergency services like towing, battery jumps, and lock-out assistance.

Financial and Operational Metrics

Operational Costs: \$350,000

Claim Costs: \$170,000

Profit Margin: 10%

Expected Customer Volume: 1,000 customers

Claim Frequency and Severity

This part of the dataset breaks down the frequency and severity of claims for each type of insurance:

Liability: 20% frequency with average severity costs of \$800.

Liability + Collision: 15% frequency with severity costs of \$3,000.

Liability + Collision + Comprehensive: 8% frequency with high severity costs of \$8,000.

This data is pivotal for InsuraMax in managing risks, allocating resources efficiently, and setting appropriate insurance premiums to maintain profitability while covering potential claims. It also aids in tailoring insurance products to meet the specific needs of their clientele, based on the historical frequency and financial impact of claims.

```
{
  "company_info": {
    "name": "InsuraMax",
    "address": {
      "street": "123 poplar st",
      "city": "Jersey City",
      "state": "NJ",
      "zip_code": "07310"
    },
    "phone_number": "908-1234",
    "email": "contact@insuramax.com",
    "website": "www.insuramax.com"
  },
  "services": [
    {
      "type": "Liability Insurance",
      "description": "Covers damages to others caused by the policyholder while operating an automobile."
    },
    {
      "type": "Collision+Liability Insurance",
      "description": "Covers damage to the policyholder's car from a collision."
    },
    {
      "type": "liability+collision+Comprehensive Insurance",
      "description": "Covers damage to the policyholder's car that is not caused by a collision."
    },
    {
      "type": "Roadside Assistance",
      "description": "Provides assistance such as towing, battery jump start, flat tire change, and lock-out assistance."
    }
  ]
}
```

```
"operational_costs": 350000,  
"claim_costs": 170000,  
"profit_margin": 0.10,  
"expected_customer_volume": 1000,  
"claim_frequency_and_severity": {  
  "liability": {  
    "frequency": 0.20,  
    "severity": 800  
  },  
  "liability+collision": {  
    "frequency": 0.15,  
    "severity": 3000  
  },  
  "liability+collision+comprehensive": {  
    "frequency": 0.08,  
    "severity": 8000  
  }  
}  
}
```



2. Description of Model

Risk assessment Model:

This risk assessment is a multifaceted Python program designed to evaluate the risk associated with insuring drivers based on their comprehensive personal, vehicular, and historical data. The model integrates various risk factors and computes an overall risk score that informs insurance premium calculations. Here's how the model operates in detail.

Model Structure

The model is composed of several functions, each responsible for assessing different aspects of risk:

Calculate Interval Factor: This function calculates the time intervals between recorded incidents (violations, accidents, suspensions) in a driver's history. Shorter intervals between incidents increase the risk score, reflecting a pattern of frequent risky behavior.

Calculate Payment History Risk: Analyzes the driver's credit score and payment timeliness. Higher credit scores and consistent on-time payments reduce the risk, whereas lower scores and missed payments increase it.

Calculate Location Risk: Assesses the risk based on the driver's location type (urban, suburban, rural), population density, and crime rate, with higher values increasing risk in more densely populated or higher-crime areas.

Calculate Age and Age Risk: Determines the driver's age from their date of birth and adjusts the risk accordingly, with younger and older drivers generally considered higher risk than middle-aged drivers.

Calculate Weighted Claims Risk: Looks at the driver's claims history, adjusting the impact of minor and total loss claims based on the number of years the driver has been insured.

Calculate Vehicle Risk: Evaluates various vehicle-related factors including age, mileage, repair history, and safety features. This function also considers the car's market value and typical repair costs to estimate financial risk associated with potential claims.

Find Carfax Report: Matches the vehicle's VIN with a corresponding Carfax report to retrieve detailed historical data essential for vehicle risk assessment.

Outputs of this model

The outputs of this model are detailed risk assessments for various drivers:

Driver Information: Includes details like name, address, driver license status, and demographic data.

Vehicle Information: Lists the vehicle's VIN, make, model, usage, mileage, and safety features.

Risk Score and Level: Numerically quantifies each driver's risk and categorizes it into levels such as Low, Moderate, or High Risk.

```
{
  {
    "driver_information": {
      "name": "John Doe",
      "address": "123 Main St, Anytown, VA, 12345",
      "date_of_birth": "1985-06-15",
      "driver_license_number": "VA123456789",
      "license_class": "Class D"
    },
    "license_status": {
      "status": "Active",
      "expiration_date": "2029-06-15"
    },
    "vehicle_information": {
      "vin": "1HGCM82633A004352",
      "make": "Honda",
      "model": "Civic",
      "year": 2018,
      "vehicle_use": "Personal",
      "annual_mileage": 15000,
      "safety_features": [
        "Automatic Emergency Braking",
        "Electronic Stability Control",
        "Adaptive Headlights"
      ]
    },
    "risk_score": 15.36,
    "risk_level": "High Risk"
  },
}
```

```
{
  {
    "driver_information": {
      "name": "Angela Thompson",
      "address": "789 Willow Ave, Arlington, VA, 78901",
      "date_of_birth": "1984-04-22",
      "driver_license_number": "VA667788990",
      "license_class": "Class D"
    },
    "license_status": {
      "status": "Active",
      "expiration_date": "2032-04-22"
    },
    "vehicle_information": {
      "vin": "1HGCV1F13JA004321",
      "make": "Honda",
      "model": "Accord",
      "year": 2018,
      "vehicle_use": "Personal",
      "annual_mileage": 5000,
      "safety_features": [
        "Automatic Emergency Braking",
        "Electronic Stability Control",
        "Adaptive Headlights"
      ]
    },
    "risk_score": 5.66,
    "risk_level": "Low Risk"
  },
}
```

```
{
  {
    "driver_information": {
      "name": "Emily Johnson",
      "address": "345 Maple Ave, Norfolk, VA, 34567",
      "date_of_birth": "1988-12-05",
      "driver_license_number": "VA112233445",
      "license_class": "Class D"
    },
    "license_status": {
      "status": "Active",
      "expiration_date": "2027-12-05"
    },
    "vehicle_information": {
      "vin": "3N1AB7AP6KY237745",
      "make": "Nissan",
      "model": "Sentra",
      "year": 2019,
      "vehicle_use": "Personal",
      "annual_mileage": 8000,
      "safety_features": [
        "Automatic Emergency Braking",
        "Electronic Stability Control"
      ]
    },
    "risk_score": 14.72,
    "risk_level": "High Risk"
  },
}
```

```
{
  {
    "driver_information": {
      "name": "Sara White",
      "address": "567 Pine St, Virginia Beach, VA, 56789",
      "date_of_birth": "1989-07-25",
      "driver_license_number": "VA998877665",
      "license_class": "Class D"
    },
    "license_status": {
      "status": "Active",
      "expiration_date": "2031-07-25"
    },
    "vehicle_information": {
      "vin": "4S4BSA9K9K3284561",
      "make": "Subaru",
      "model": "Outback",
      "year": 2019,
      "vehicle_use": "Personal",
      "annual_mileage": 14000,
      "safety_features": [
        "Automatic Emergency Braking",
        "Electronic Stability Control",
        "Adaptive Headlights"
      ]
    },
    "risk_score": 12.06,
    "risk_level": "Moderate Risk"
  },
}
```

Base Price Calculation:

Model Overview

The model uses insurance company's financials, service details, and insurance claim data to compute the base price for each insurance service offered. Company.json is given as input to this model.

Inputs:

The model takes the following inputs:

Operational Costs: The fixed costs associated with running the insurance operations.

Claim Costs: The expected costs from claims that have been, or are projected to be, paid out.

Profit Margin: The desired margin that the insurance company aims to achieve for profitability.

Expected Customer Volume: The total number of customers the insurance company expects to serve.

Claim Frequency and Severity: Data indicating how often claims occur (frequency) and the average cost of these claims (severity) for different types of insurance services.

Steps in the Model:

Calculate Total Costs: Adds operational and claim costs to determine the total expenses the company needs to cover.

Calculate Required Revenue: Adjusts the total costs by the profit margin to determine how much revenue is needed to cover costs while achieving the desired profit. Required Revenue equals Total Costs divided by the difference between 1 and the Profit Margin.

Determine Base Price Per Policy: Divides the required revenue by the expected customer volume to find the average base price per policy before considering specific claim risks.

Adjust Prices for Each Insurance Type: For each type of insurance offered, the base price is adjusted to account for the expected costs due to claims. This is calculated by adding the product of claim frequency and severity to the base price. This step ensures that policies with higher expected claim costs are priced accordingly to maintain profitability.

Monthly Pricing: Converts the adjusted annual price to a monthly rate to provide a practical figure that can be communicated to customers.

Outputs:

The output of the model is a set of monthly prices for each type of insurance service. These prices are specifically adjusted to cover all associated costs (operational and claim-related) and ensure the company

reaches its profit goals. This strategic pricing helps manage financial risk and tailors prices to the specific risks associated with different types of insurance policies.

Premium Evaluating Model:

The premium evaluating model functions as an advanced decision-support tool in an insurance company's pricing strategy, effectively determining optimal premiums for drivers by integrating input from two crucial components: the risk assessment model and the base price evaluation model. Here's how the model operates:

Input Integration

Risk Score: Received from the risk assessment model, this score quantifies the likelihood and potential severity of claims based on the driver's historical data, vehicle information, and behavior patterns. The higher the risk score, the greater the potential cost to the insurer, which is reflected in higher premiums.

Base Price: Derived from the base price evaluation model, this figure represents the initial calculation of insurance costs without considering individual driver risk. It accounts for general factors like operational costs, claim costs, expected customer volume, and desired profit margins.

Premium Calculation

Risk Adjustment: The model adjusts the base price according to the risk score. Specific multipliers are applied to the base price depending on the risk category (low, moderate, high, very high) that the risk score falls into. This adjustment ensures that the premium reflects the actual risk associated with insuring the driver.

Discount Evaluation: The model evaluates eligibility for discounts using the `all` function from the `dgalPy` library, based on criteria such as safe driving points, annual mileage, and the number of safety features in the vehicle. These discounts are designed to incentivize safe driving behaviors and the use of safer vehicles, reducing the overall risk and thus the premium cost. By ensuring that all specified conditions are met, the `all` function provides a rigorous method for applying discounts, enhancing the precision of the model's risk adjustments and premium calculations.

Premium Finalization

Monthly Premiums: After incorporating risk-related adjustments and applicable discounts, the model calculates the final monthly premiums for each type of insurance coverage. This ensures that the premiums are not only competitive and sufficient to cover anticipated claim costs but also fair in reflecting the individual driver's risk profile.

Output

The model outputs the detailed premium structure for each insurance type the driver may consider, indicating the final premium amounts and whether any discounts have been applied. This output provides transparent and customized pricing information that helps both the insurance provider in risk management and the customer in understanding their insurance costs.

Overall, the premium calculation model leverages sophisticated calculations to align insurance premiums closely with assessed risk, ensuring that pricing strategies are both data-driven and dynamically responsive to individual risk profiles.

```
{
  "name": "Jane Smith",
  "address": "234 Elm St, Richmond, VA, 23456",
  "date_of_birth": "1992-08-20",
  "driver_license_number": "VA987654321",
  "license_class": "Class D",
  "vehicle_information": {
    "vin": "2T1BURHE8JC015872",
    "make": "Toyota",
    "model": "Corolla",
    "year": 2020,
    "vehicle_use": "Personal",
    "annual_mileage": 10000,
    "safety_features": [
      "Automatic Emergency Braking",
      "Electronic Stability Control",
      "Adaptive Headlights"
    ]
  },
  "risk_score": 13.62,
  "premiums": {
    "premiums": {
      "liability": "$83.80",
      "liability+collision": "$107.54",
      "liability+collision+comprehensive": "$130.15"
    },
    "discount_applied": false
  },
},
{
  "name": "Michael Lee",
  "address": "456 Oak Rd, Fairfax, VA, 45678",
  "date_of_birth": "1974-03-30",
  "driver_license_number": "VA556677889",
  "license_class": "Class D",
  "vehicle_information": {
    "vin": "3FA6P0K92HR294653",
    "make": "Ford",
    "model": "Fusion",
    "year": 2017,
    "vehicle_use": "Personal",
    "annual_mileage": 9900,
    "safety_features": [
      "Automatic Emergency Braking",
      "Electronic Stability Control",
      "Adaptive Headlights"
    ]
  },
  "risk_score": 5.96,
  "premiums": {
    "premiums": {
      "liability": "$63.63",
      "liability+collision": "$82.44",
      "liability+collision+comprehensive": "$99.78"
    },
    "discount_applied": true
  },
},
}
```

```

{
  Click to collapse the range.  talie Martinez",
  "address": "901 Pine St, Lynchburg, VA, 90123",
  "date_of_birth": "1982-02-28",
  "driver_license_number": "VA334455667",
  "license_class": "Class D",
  "vehicle_information": {
    "vin": "1N4BL4CV6LC125678",
    "make": "Nissan",
    "model": "Altima",
    "year": 2020,
    "vehicle_use": "Personal",
    "annual_mileage": 12000,
    "safety_features": [
      "Automatic Emergency Braking",
      "Electronic Stability Control",
      "Adaptive Headlights"
    ]
  },
  "risk_score": 9.26,
  "premiums": {
    "premiums": {
      "liability": "$70.70",
      "liability+collision": "$90.50",
      "liability+collision+comprehensive": "$116.70"
    },
    "discount_applied": false
  }
},

```

```

{
  {
    "name": "John Doe",
    "address": "123 Main St, Anytown, VA, 12345",
    "date_of_birth": "1985-06-15",
    "driver_license_number": "VA123456789",
    "license_class": "Class D",
    "vehicle_information": {
      "vin": "1HGM082633A004352",
      "make": "Honda",
      "model": "Civic",
      "year": 2018,
      "vehicle_use": "Personal",
      "annual_mileage": 15000,
      "safety_features": [
        "Automatic Emergency Braking",
        "Electronic Stability Control",
        "Adaptive Headlights"
      ]
    },
    "risk_score": 15.36,
    "premiums": {
      "premiums": {
        "liability": "$95.30",
        "liability+collision": "$132.75",
        "liability+collision+comprehensive": "$157.30"
      },
      "discount_applied": false
    }
  },

```

Conclusion

Throughout this project, we have developed and refined a comprehensive auto insurance recommendation system designed for modern insurance companies. The system integrates multiple models that analyze a vast array of data points from driver history, vehicle information, and financial metrics to assess risk and determine insurance premiums accurately.

The risk assessment model serves as the cornerstone of this system, leveraging detailed data to calculate individual risk scores based on drivers' behavior, vehicle condition, and historical claims. This model not only assesses risk with high precision but also categorizes it into actionable segments that directly influence premium calculations.

Following the risk evaluation, the base price evaluation model determines the foundational costs of insurance policies. This model ensures that the operational and claim costs are covered while achieving the desired profit margins, setting the stage for the final premium calculations.

The premium calculation model then takes these base prices and adjusts them according to the calculated risk scores and potential discounts. The use of advanced functions, such as the `all` function from the `dgalPy` library, ensures that discounts are applied only when stringent criteria are met, thus rewarding safe driving behaviors and the use of vehicles with advanced safety features. This method ensures that premiums are not only fair and reflective of the actual risk but also competitive in the market.

This system enhances the insurance company's ability to offer dynamic pricing, which is more aligned with individual risk factors and market conditions. By doing so, it supports the company in achieving financial stability and customer satisfaction through tailored insurance offerings that incentivize lower risk and promote safety on the roads.

In conclusion, this project has successfully created a robust framework that can adapt to various data inputs and conditions, providing a scalable solution for auto insurance companies aiming to modernize their pricing strategies and improve their risk management practices.

Future work:

Looking ahead, the next phase of this project will focus on enhancing the auto insurance recommendation system by implementing dynamic decision-making capabilities, primarily through the continuous integration of data. This approach aims to refine risk assessments and pricing strategies continually, ensuring that the insurance company can maintain competitive and fair pricing.

Future developments will involve setting up mechanisms to gather and integrate a wide array of data continuously, including real-time driving behaviors, updates on vehicle conditions, and changes in legal regulations. This continuous data integration will allow the system to stay current with the latest information, enabling timely adjustments to risk assessments and premium calculations.

We plan to leverage machine learning algorithms to develop adaptive pricing models that automatically adjust to changing risk profiles and market conditions. This proactive approach will allow the insurance company to respond swiftly to market trends and data insights, maintaining its competitive edge.

Additionally, there is potential to enhance personalization of insurance offers, tailoring them more closely to individual customer preferences and needs. This could include developing predictive models to anticipate changes in customer circumstances and offering customized insurance solutions preemptively.

Integration with Internet of Things (IoT) devices, such as connected vehicles and telematics systems, will also be explored. These devices can provide real-time insights into vehicle usage patterns and driver behavior, which are crucial for dynamically adjusting premiums and managing risks more effectively.

To support these advancements, we will also look into applying more sophisticated risk modeling techniques that can handle the complexity and variability of modern driving environments and driver behaviors, enhancing the accuracy of risk assessments. Moreover, developing an automated claims processing module that integrates with the risk assessment system will streamline claims management, reduce processing times, and decrease the likelihood of fraud.

By advancing these areas, the auto insurance recommendation system will evolve into a more dynamic, efficient, and customer-centric solution, improving customer satisfaction through more tailored and responsive insurance services while bolstering the company's market position.